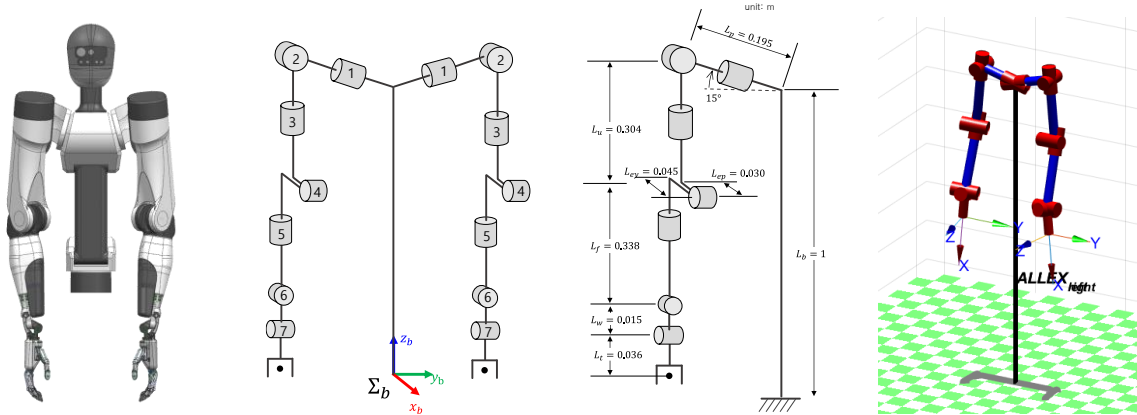


Compliant Robotics Assignment 2

2024.10.22

The following figures show the 7-DOF arms of the humanoid robot LIMS-ALEX. Answer the following questions.



1. Make a MATLAB Function J_SerialLink.m calculating Jacobian of manipulators. For the kinematics calculation, use "mdl_LIMS_ALEX_Simple." The input and output are as follows:

Function J = J_SerialLink(T)

T : 4x4x9 matrix of HT, $T(:,1)= {}^bT_o$, $T(:,2)= {}^oT_1$, ..., $T(:,8)= {}^6T_7$, $T(:,9)= {}^7T_t$

J : 6x7 Jacobian matrix

2. Calculate the Jacobians of the following poses and compare them with the result of Jacobian using Robotics Toolbox.

Display velocity manipulability ellipsoid and the force manipulability ellipsoid of the right arm of the following poses using Robotics Toolbox.

Pose 1: left arm [0, 0, 0, 0, 0, 0, 0], right arm [0, 0, 0, 0, 0, 0, 0]

pose 2: left arm [0, $\pi/4$, 0, $-\pi/2$, 0, 0, $-\pi/4$], right arm [0, $-\pi/4$, 0, $-\pi/2$, 0, 0, $-\pi/4$]

pose 3: left arm [$-\pi/3$, $\pi/6$, $-\pi/4$, $-\pi/2$, 0, $-\pi/8$, 0], right arm [$\pi/3$, $-\pi/6$, $\pi/4$, $-\pi/2$, 0, $\pi/8$, 0]